

FIND KEY AND OPEN DOOR WITH DQN

Introduction:

The assignment is about evaluating a DQN model that can solve (find key and open door) problem. In this report I will talk about the main lines I had to implement in the notebook then I will have discussion about my experience in this assignment.

Update target network:

The `soft\_update` function is designed to perform a soft update of the parameters of the target network in a Deep Q-Network (DQN) setting. This function updates the target network's parameters to slowly track the parameters of the local network using the formula:

$$\theta_{\text{target}} = \tau * \theta_{\text{local}} + (1 - \tau) * \theta_{\text{target}}$$

This soft update mechanism helps in stabilizing the training process by preventing large oscillations in Q-value estimates, thereby enabling more stable learning. The function iterates through the parameters of both the local and target models, and updates each parameter of the target model to be a blend of its current value and the corresponding parameter of the local model.

DQN training loop:

This code trains a Deep Q-Network (DQN) agent to play Pong. Initially, variables for returns, losses, and a replay buffer are set up. The DQN and target network, along with the timesteps, are loaded from a checkpoint. The optimizer and loss function are then initialized.

The training loop begins by resetting the environment for each episode and initializing the return (`ret`) and `done` flag. An epsilon-greedy policy is used for action selection, where actions are chosen either randomly or based on the current Q-values predicted by the DQN. The chosen action is executed in the environment, and the next state and reward are observed. This transition is stored in the replay buffer.

When enough experiences are available in the replay buffer, a minibatch is sampled to update the policy. Q-values for the current and next states are computed, and target Q-values are calculated using the rewards and the maximum Q-values of the next states from the target network. The loss between the predicted and target Q-values is computed, backpropagated, and used to update the DQN network's parameters. Additionally, a soft update of the target network is performed to ensure stable learning. Throughout the training process, episode returns are tracked and printed. Checkpoints are saved periodically to preserve the agent's progress. This systematic approach allows the DQN to learn from its interactions with the environment, gradually improving its policy over time.

Discussion:

First I have trained the model on the (MinigridEmpty5x5ImgObs) environment and it after some computations it gave me very good average score of 0.9 without editing the hyperparameters. But the problem was that I had to change the sequence of the loop as I had to make the Q-value computation before the computing the target.

Second I started training on the (DoorKey6x6ImgObs) environment, and it took too many iterations to see a little difference in the performance and the score started with zero then during training loops it started to raise exponentially and very slow till it reaches to 0.5 then it drops very hard till it reaches to zero again. So to solve this edited the batch_size to 128 then 256 then return back to 5 that by visualizing each training session also edited the num_episodes to 1500 it made some difference in the first sessions also epsilon_lb and learning rate they sometimes make a little difference.

After many training sessions and blocking 6 google accounts by Colab due to GPU limit, I finally reached to an average score of 0.67 but when submitting on the server it gave me zero average and do not know why, also worth mentioning that after making one more training session to improve the score it dropped from 0.67 to 0.33 also nor sure why.

In conclusion there was some issues in implementing the training loop but it was not that hard also training on (MinigridEmpty5x5ImgObs) environment was very sensible and smooth, but training on (DoorKey6x6ImgObs) environment was hard and took so much time to get a good result. I enjoyed that assignment but it is very depressive that the server did not work.